Ansible -01

Task : 1

1.Watch ansible-01 video and write down notes.

Task : 2

2.Setup one master and two worker nodes in ansible.

Setup 1 Ansible Master (Control Node) + 2 Worker Nodes (Ubuntu)

All 3 machines = Ubuntu

—> Architecture (remember this)

- **Master** → Ansible installed
- **Workers (2)** → No Ansible
- Communication → **SSH**
- Type → **Agentless**

```
Control Node (Ubuntu)
        |
       SSH
   ----------
   |         |
Worker1    Worker2
```

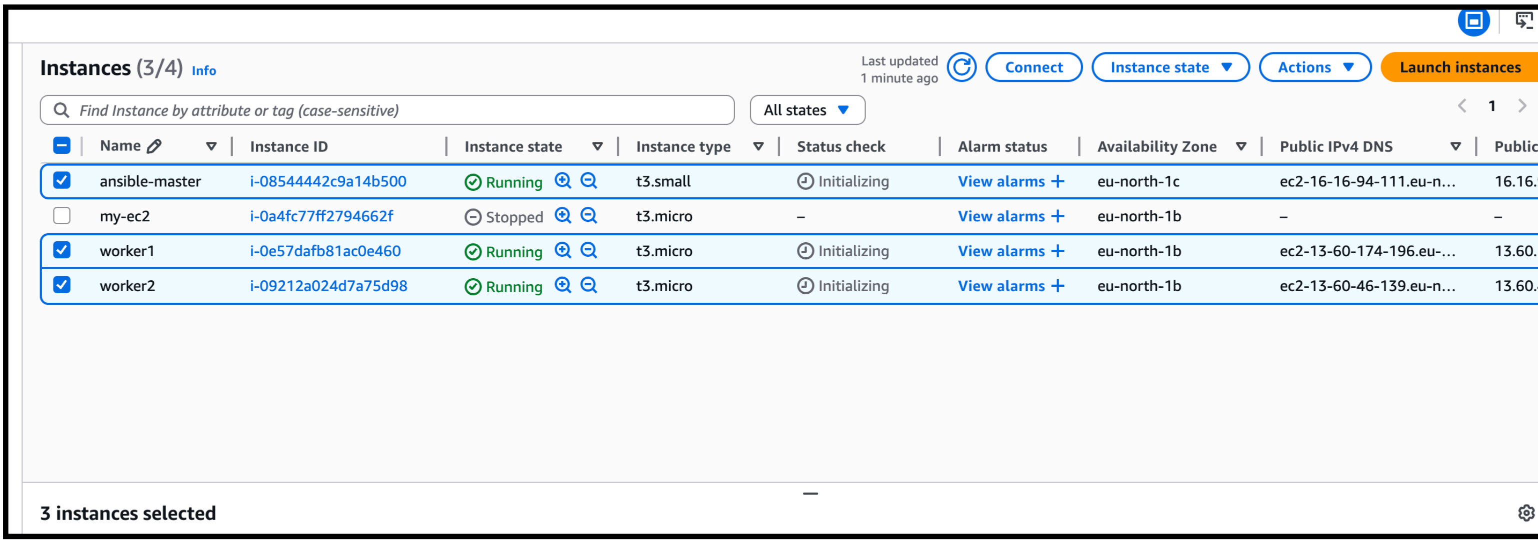
—> launch three instances

—> 2️⃣ Create 3 EC2 instances (AWS example)

Node	OS	Purpose
Control	Amazon Linux / Ubuntu	Ansible installed
Worker-1	Amazon Linux / Ubuntu	Managed node
Worker-2	Amazon Linux / Ubuntu	Managed node

✔ Same VPC
✔ Same key-pair
✔ Security Group: allow **SSH (22)**

—> I used ubuntu OS



Step-by-Step Setup: Ansible with 1 Master + 2 Workers (Ubuntu EC2)

Step 1:

—->go to your ansible master server

Check:

```
python3 --version
```

```
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.14.0-1018-aws x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

 System information as of Wed Jan 28 09:11:17 UTC 2026

  System load:  0.0               Temperature:           -273.1 C
  Usage of /:   38.5% of 6.71GB   Processes:             115
  Memory usage: 11%               Users logged in:       0
  Swap usage:   0%                IPv4 address for ens5: 172.31.2.72


Expanded Security Maintenance for Applications is not enabled.

60 updates can be applied immediately.
32 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status


Last login: Wed Jan 28 07:25:17 2026 from 13.48.4.202
ubuntu@ip-172-31-2-72:~$ sudo -i
root@ip-172-31-2-72:~# python3 --version
Python 3.12.3
root@ip-172-31-2-72:~#
```

—-> sudo apt update -y

—-> sudo apt install ansible –y

```
root@ip-172-31-2-72:~# python3 --version
Python 3.12.3
root@ip-172-31-2-72:~# apt install ansible -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  ansible-core python3-argcomplete python3-dnspython python3-kerberos python3-libcloud python3-lockfile python
  python3-selinux python3-simplejson python3-winrm python3-xmltodict
Suggested packages:
  cowsay sshpass python3-trio python3-aioquic python3-h2 python3-httpx python3-httpcore python-lockfile-doc
The following NEW packages will be installed:
  ansible ansible-core python3-argcomplete python3-dnspython python3-kerberos python3-libcloud python3-lockfil
  python3-selinux python3-simplejson python3-winrm python3-xmltodict
0 upgraded, 15 newly installed, 0 to remove and 62 not upgraded.
Need to get 19.5 MB of archives.
After this operation, 315 MB of additional disk space will be used.
Get:1 http://eu-north-1.ec2.archive.ubuntu.com/ubuntu noble/universe amd64 python3-resolvelib all 1.0.1-1 [25.
Get:2 http://eu-north-1.ec2.archive.ubuntu.com/ubuntu noble/main amd64 python3-dnspython all 2.6.1-1ubuntu1 [1
Get:3 http://eu-north-1.ec2.archive.ubuntu.com/ubuntu noble/universe amd64 ansible-core all 2.16.3-0ubuntu2 [1
Get:4 http://eu-north-1.ec2.archive.ubuntu.com/ubuntu noble/universe amd64 ansible all 9.2.0+dfsg-0ubuntu5 [16
```

—->ansible - -version

```
root@ip-172-31-2-72:~# ansible --version
ansible [core 2.16.3]
  config file = None
  configured module search path = ['/root/.ansible/plugins/modules', '/usr/share/ansible/plugins/modules']
  ansible python module location = /usr/lib/python3/dist-packages/ansible
  ansible collection location = /root/.ansible/collections:/usr/share/ansible/collections
  executable location = /usr/bin/ansible
  python version = 3.12.3 (main, Nov  6 2025, 13:44:16) [GCC 13.3.0] (/usr/bin/python3)
  jinja version = 3.1.2
  libyaml = True
root@ip-172-31-2-72:~#
```

Step: 2

Here we have no cfg file by default

By default, **Ubuntu does not create /etc/ansible/ansible.cfg or /etc/ansible/hosts anymore** — instead, you're expected to **create your own config and inventory** in your project directory.

—>sudo mkdir -p /etc/ansible

—-> sudo vi /etc/ansible/ansible.cfg

```
root@ip-172-31-2-72:~# sudo mkdir -p /etc/ansible
root@ip-172-31-2-72:~# cd /etc/ansible/
root@ip-172-31-2-72:/etc/ansible# ls
root@ip-172-31-2-72:/etc/ansible# vi /etc/ansible/ansible.cfg
root@ip-172-31-2-72:/etc/ansible# ls
ansible.cfg
```

vi /etc/ansible/ansible.cfg

```
[defaults]
remote_user = root
host_key_checking = False
retry_files_enabled = False
forks = 10

~
~
~
~
~
```

—> save :wq!

Created global inventory

```
sudo vi /etc/ansible/hosts
```

```
[all]
172.31.33.34
172.31.37.159

[worker1]
172.31.33.34

[worker2]
172.31.37.159

~
~
~
~
~
~
~
~
~
~
```

Save and exit

—-> now check the
—> ansible all -m ping

```
root@ip-172-31-2-72:~# ls
snap
root@ip-172-31-2-72:~# cd snap/
root@ip-172-31-2-72:~/snap# ls
amazon-ssm-agent
root@ip-172-31-2-72:~/snap# cd
root@ip-172-31-2-72:~# ansible all -m ping
172.31.37.159 | UNREACHABLE! => {
    "changed": false,
    "msg": "Failed to connect to the host via ssh: Warning: Permanently added '172.31.37.159' (ED25519) to the list of known hosts.\r\nroot@172.31.37.159: Permission de
    "unreachable": true
}
172.31.33.34 | UNREACHABLE! => {
    "changed": false,
    "msg": "Failed to connect to the host via ssh: Warning: Permanently added '172.31.33.34' (ED25519) to the list of known hosts.\r\nroot@172.31.33.34: Permission deni
    "unreachable": true
}
root@ip-172-31-2-72:~# ssh -keygen
Bad escape character 'ygen'
```

—-> we are able to see error because we haven’t ping (connectivity to the other instances)

Step :3

—> to have connectivity we need to create a ssh-keygen in ansible master server

means Ansible is trying to connect with the ubuntu user, but the SSH key is missing or not set up properly between your master (control node) and the worker nodes.

Fix SSH Access Between Master and Workers

Do ssh-kegen from Master server

```
root@ip-172-31-2-72:~# ssh-keygen
Generating public/private ed25519 key pair.
Enter file in which to save the key (/root/.ssh/id_ed25519):
Enter passphrase (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /root/.ssh/id_ed25519
Your public key has been saved in /root/.ssh/id_ed25519.pub
The key fingerprint is:
SHA256:kq2gYOhrJy2ZkzrKJoFzA5WhP+KC4DMF90fs0lG5PjA root@ip-172-31-2-72
The key's randomart image is:
+--[ED25519 256]--+
|   .o      ..    |
|  .o    . ..     |
|.o .    +  .     |
|o.o . +E..       |
|*oo..oo+S        |
|@o+o .oo o       |
|oX*.   .    .    |
|=X+o             |
|O+=              |
+----[SHA256]-----+
root@ip-172-31-2-72:~# cat /root/.ssh/id_ed25519.pub
ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAID5LTgnUy2XDhyWRs5M3VNkekezuSun+bD+qqnUp87xT root@ip-172-31-2-72
root@ip-172-31-2-72:~# ansible all -m ping
```

—> now after creating key copy the pub key

cat /root/.ssh/id_ed25519.pub

—> copy the key and paste in other two instances —>.ssh/authorized_keys

—> open worker1 server

```
root@ip-172-31-33-34:~# cd .ssh
root@ip-172-31-33-34:~/.ssh# ls
authorized_keys
root@ip-172-31-33-34:~/.ssh# vi authorized_keys
root@ip-172-31-33-34:~/.ssh#
```

i-0e57dafb81ac0e460 (worker1)

PublicIPs: 13.60.174.196   PrivateIPs: 172.31.33.34


```
ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAID5LTgnUy2XDhyWRs5M3VNkekezuSun+bD+qqnUp87xT root@ip-172-31-2-72

no-port-forwarding,no-agent-forwarding,no-X11-forwarding,command="echo 'Please login as the user \"ubuntu\" rather than the user \"root\".';echo;sleep 10;exit 142" ssh-rsa AAAAB3NzaC1yc2
AAADAQABAAABAQDWNeXm5oAOnWRicv1796/qbv0Z17yuG7Pm0hEG7ZGLvOWroKzibsWmQcCF02wT1Xlrh3R2pIdJv/SdkpbHJNUIbPpSE0OHYS+n2/MRki35CS2nVH5usGKmvJM/Ykhh6cyRdPPUYvPNm1ADvxshsLpxI3P7kBh9Wk8hvA8P0diwPD
PuyPxhE3qSN2RXeLmHSr1pb4tM+ed8RwR/QDSSpmH213Siyr/+U4u3MXikWAC3GI8GWPm3oew5x3PVW4KtjAxbeB5tIQP8gedeGtfGgq0Lv+sJVV2u6ELUhGeeYXAwFTXP2WrhhD/HRjiFLlmNQAIQyt+E2GEIugx1TWUnfz ansible
```


Similarly open worker2 server and add same key in authorizd_keys

```
62 packages can be upgraded. Run 'apt list --upgradable' to see th
root@ip-172-31-37-159:~# cd .ssh
root@ip-172-31-37-159:~/.ssh# vi authorized_keys
root@ip-172-31-37-159:~/.ssh#
```

i-09212a024d7a75d98 (worker2)

PublicIPs: 13.60.46.139   PrivateIPs: 172.31.37.159

Step 5:

—> now check in ansible master server
—>ansible all -m ping

```
root@ip-172-31-2-72:~# ansible all -m ping
172.31.33.34 | SUCCESS => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3"
    },
    "changed": false,
    "ping": "pong"
}
172.31.37.159 | SUCCESS => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3"
    },
    "changed": false,
    "ping": "pong"
}
root@ip-172-31-2-72:~#
```

i-08544442c9a14b500 (ansible-master)

PublicIPs: 16.16.94.111 PrivateIPs: 172.31.2.72

Task 3:
3.Execute the adhoc command shared in #dvps-cloud-documents-16 Channel.

To check the connetivity between slaves
ansible all -m ping

```
root@ip-172-31-2-72:/etc/ansible# ansible all -m ping
172.31.33.34 | SUCCESS => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3"
    },
    "changed": false,
    "ping": "pong"
}
172.31.37.159 | SUCCESS => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3"
    },
    "changed": false,
    "ping": "pong"
}
root@ip-172-31-2-72:/etc/ansible#
```

ansible worker1 -m ping

```
root@ip-172-31-2-72:/etc/ansible# ansible worker1 -m ping
172.31.33.34 | SUCCESS => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3"
    },
    "changed": false,
    "ping": "pong"
}
root@ip-172-31-2-72:/etc/ansible#
```

—> because we have given group names worker1, worker2

```
[all]
172.31.33.34
172.31.37.159

[worker1]
172.31.33.34

[worker2]
172.31.37.159

~
~
~
~
~
~
~
~
~
~
```

How to gather facts of slave machine
ansible all -m setup

```
root@ip-172-31-2-72:~# ansible all -m setup
172.31.37.159 | SUCCESS => {
    "ansible_facts": {
        "ansible_all_ipv4_addresses": [
            "172.31.37.159"
        ],
        "ansible_all_ipv6_addresses": [
            "fe80::879:3eff:febc:c2d5"
        ],
        "ansible_apparmor": {
            "status": "enabled"
        },
        "ansible_architecture": "x86_64",
        "ansible_bios_date": "10/16/2017",
        "ansible_bios_vendor": "Amazon EC2",
        "ansible_bios_version": "1.0",
        "ansible_board_asset_tag": "i-09212a024d7a75d98",
        "ansible_board_name": "NA",
        "ansible_board_serial": "NA",
        "ansible_board_vendor": "Amazon EC2",
        "ansible_board_version": "NA",
        "ansible_chassis_asset_tag": "Amazon EC2",
        "ansible_chassis_serial": "NA",
        "ansible_chassis_vendor": "Amazon EC2",
        "ansible_chassis_version": "NA",
        "ansible_cmdline": {
            "BOOT_IMAGE": "/vmlinuz-6.14.0-1018-aws",
            "console": "ttyS0",
            "nvme_core.io_timeout": "4294967295",
            "panic": "-1",
            "ro": true,
            "root": "PARTUUID=2baf9194-a5fb-4b12-adb0-45166f0feb06"
        },
        "ansible_date_time": {
            "date": "2026-01-28",
            "day": "28",
```

TO checkthe uptime of a slave machine
ansible all -a uptime

```
root@ip-172-31-2-72:~# ansible all -a uptime
172.31.33.34 | CHANGED | rc=0 >>
 10:22:32 up  1:10,  2 users,  load average: 0.00, 0.00, 0.00
172.31.37.159 | CHANGED | rc=0 >>
 10:22:32 up  1:10,  2 users,  load average: 0.00, 0.00, 0.00
root@ip-172-31-2-72:~#
```

Ansible all -m ping -vvvv

—-> to check with all log details

```
root@ip-172-31-2-72:~# ansible all -m ping -vvvv
ansible [core 2.16.3]
  config file = /etc/ansible/ansible.cfg
  configured module search path = ['/root/.ansible/plugins/modules', '/usr/share/ansible/plugins/modules']
  ansible python module location = /usr/lib/python3/dist-packages/ansible
  ansible collection location = /root/.ansible/collections:/usr/share/ansible/collections
  executable location = /usr/bin/ansible
  python version = 3.12.3 (main, Nov  6 2025, 13:44:16) [GCC 13.3.0] (/usr/bin/python3)
  jinja version = 3.1.2
  libyaml = True
Using /etc/ansible/ansible.cfg as config file
setting up inventory plugins
Loading collection ansible.builtin from
host_list declined parsing /etc/ansible/hosts as it did not pass its verify_file() method
script declined parsing /etc/ansible/hosts as it did not pass its verify_file() method
auto declined parsing /etc/ansible/hosts as it did not pass its verify_file() method
Parsed /etc/ansible/hosts inventory source with ini plugin
Loading callback plugin minimal of type stdout, v2.0 from /usr/lib/python3/dist-packages/ansible/plugins/callback/minimal.py
Skipping callback 'default', as we already have a stdout callback.
Skipping callback 'minimal', as we already have a stdout callback.
Skipping callback 'oneline', as we already have a stdout callback.
<172.31.33.34> ESTABLISH SSH CONNECTION FOR USER: root
<172.31.33.34> SSH: EXEC ssh -vvv -C -o ControlMaster=auto -o ControlPersist=60s -o StrictHostKeyChecking=no -o KbdInteractiveAuthentication=no -o Pre
t=10 -o 'ControlPath="/root/.ansible/cp/9291b3cf39"' 172.31.33.34 '/bin/sh -c '"'"'echo ~root && sleep 0'"'"''
<172.31.37.159> ESTABLISH SSH CONNECTION FOR USER: root
<172.31.37.159> SSH: EXEC ssh -vvv -C -o ControlMaster=auto -o ControlPersist=60s -o StrictHostKeyChecking=no -o KbdInteractiveAuthentication=no -o Pr
ut=10 -o 'ControlPath="/root/.ansible/cp/7acb44ebe4"' 172.31.37.159 '/bin/sh -c '"'"'echo ~root && sleep 0'"'"''
<172.31.33.34> (0, b'/root\n', b"OpenSSH_9.6p1 Ubuntu-3ubuntu13.14, OpenSSL 3.0.13 30 Jan 2024\r\ndebug1: Reading configuration data /etc/ssh/ssh_conf
21: Applying options for *\r\ndebug2: resolve_canonicalize: hostname 172.31.33.34 is address\r\ndebug3: expanded UserKnownHostsFile '~/.ssh/known_host
uto-mux: Trying existing master at '/root/.ansible/cp/9291b3cf39'\r\ndebug2: fd 3 setting O_NONBLOCK\r\ndebug2: mux_client_hello_exchange: master vers
3: mux_client_request_alive: entering\r\ndebug3: mux_client_request_alive: done pid = 2396\r\ndebug3: mux_client_request_session: session request sent
```

```
x_client_request_alive: entering\r\ndebug3: mux_client_request_alive: done pid = 239
ndebug2: Received exit status from master 0\r\n")
<172.31.33.34> (0, b'', b"OpenSSH_9.6p1 Ubuntu-3ubuntu13.14, OpenSSL 3.0.13 30 Jan 2
lying options for *\r\ndebug2: resolve_canonicalize: hostname 172.31.33.34 is addres
: Trying existing master at '/root/.ansible/cp/9291b3cf39'\r\ndebug2: fd 3 setting O_
client_request_alive: entering\r\ndebug3: mux_client_request_alive: done pid = 2396\
ebug2: Received exit status from master 0\r\n")
172.31.33.34 | SUCCESS => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3"
    },
    "changed": false,
    "invocation": {
        "module_args": {
            "data": "pong"
        }
    },
    "ping": "pong"
}
172.31.37.159 | SUCCESS => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3"
    },
    "changed": false,
    "invocation": {
        "module_args": {
            "data": "pong"
        }
    },
    "ping": "pong"
}
root@ip-172-31-2-72:~#
```

check the free memory or memory usage of host
ansible all -a "free -m"

```
root@ip-172-31-2-72:~# ansible all -a "free -m"
172.31.33.34 | CHANGED | rc=0 >>
              total        used        free      shared  buff/cache   available
Mem:            914         384         173           2         517         529
Swap:             0           0           0
172.31.37.159 | CHANGED | rc=0 >>
              total        used        free      shared  buff/cache   available
Mem:            914         382          92           2         601         531
Swap:             0           0           0
root@ip-172-31-2-72:~#
```

Create directory using shell module

Ansible all -m shell -a “mkdir ansible.log”

```
root@ip-172-31-2-72:/etc/ansible# ansible all -m shell -a "mkdir ansible.log"
172.31.33.34 | CHANGED | rc=0 >>

172.31.37.159 | CHANGED | rc=0 >>

root@ip-172-31-2-72:/etc/ansible#
```

—> what it does it will create ansible.log directory in all workers server

```
root@ip-172-31-33-34:~# ls
ansible.log  snap
root@ip-172-31-33-34:~#
```

i-0e57dafb81ac0e460 (worker1)

PublicIPs: 13.60.174.196   PrivateIPs: 172.31.33.34

```
root@ip-172-31-37-159:/# cd
root@ip-172-31-37-159:~# ls
ansible.log  snap
root@ip-172-31-37-159:~#
```

i-09212a024d7a75d98 (worker2)

PublicIPs: 13.60.46.139 PrivateIPs: 172.31.37.159

Install a package using yum command
ansible all -m yum -a "name=httpd state=installed"

—> iam using ubuntu
So ansible all -m apt -a 'name=apache2 state=latest'

```
root@ip-172-31-2-72:/etc/ansible# ansible all -m apt -a 'name=apache2 state=latest'
172.31.33.34 | CHANGED => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3"
    },
    "cache_update_time": 1769593307,
    "cache_updated": false,
    "changed": true,
    "stderr": "",
    "stderr_lines": [],
    "stdout": "Reading package lists...\nBuilding dependency tree...\nReading state information...\nThe following additional pac
-utils libapr1t64 libaprutil1-dbd-sqlite3\n  libaprutil1-ldap libaprutil1t64 liblua5.4-0 ssl-cert\nSuggested packages:\n  apache
owser\nThe following NEW packages will be installed:\n  apache2 apache2-bin apache2-data apache2-utils libapr1t64\n  libaprutil1
-cert\n0 upgraded, 10 newly installed, 0 to remove and 62 not upgraded.\nNeed to get 2090 kB of archives.\nAfter this operation,
//eu-north-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 libapr1t64 amd64 1.7.2-3.1ubuntu0.1 [108 kB]\nGet:2 http://e
aprutil1t64 amd64 1.6.3-1.1ubuntu7 [91.9 kB]\nGet:3 http://eu-north-1.ec2.archive.ubuntu.com/ubuntu noble/main amd64 libaprutil1
eu-north-1.ec2.archive.ubuntu.com/ubuntu noble/main amd64 libaprutil1-ldap amd64 1.6.3-1.1ubuntu7 [9116 B]\nGet:5 http://eu-nort
-0 amd64 5.4.6-3build2 [166 kB]\nGet:6 http://eu-north-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 apache2-bin amd6
archive.ubuntu.com/ubuntu noble-updates/main amd64 apache2-data all 2.4.58-1ubuntu8.10 [163 kB]\nGet:8 http://eu-north-1.ec2.arc
s amd64 2.4.58-1ubuntu8.10 [98.1 kB]\nGet:9 http://eu-north-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 apache2 amd
2.archive.ubuntu.com/ubuntu noble/main amd64 ssl-cert all 1.1.2ubuntu1 [17.8 kB]\nPreconfiguring packages ...\nFetched 2090 kB i
bapr1t64:amd64.\r\n(Reading database ... \r(Reading database ... 5%\r(Reading database ... 10%\r(Reading database ... 15%\r(Read
tabase ... 30%\r(Reading database ... 35%\r(Reading database ... 40%\r(Reading database ... 45%\r(Reading database ... 50%\r(Rea
atabase ... 65%\r(Reading database ... 70%\r(Reading database ... 75%\r(Reading database ... 80%\r(Reading database ... 85%\r(Re
database ... 100%\r(Reading database ... 71752 files and directories currently installed.)\r\nPreparing to unpack .../0-libapr1t
amd64 (1.7.2-3.1ubuntu0.1) ...\r\nSelecting previously unselected package libaprutil1t64:amd64.\r\nPreparing to unpack .../1-lib
rutil1t64:amd64 (1.6.3-1.1ubuntu7) ...\r\nSelecting previously unselected package libaprutil1-dbd-sqlite3:amd64.\r\nPreparing t
```

i-08544442c9a14b500 (ansible-master)

PublicIPs: 16.16.94.111 PrivateIPs: 172.31.2.72

CloudShell Feedback Console Mobile App

—-> now check your other workers servers if apache2 package is installed or not

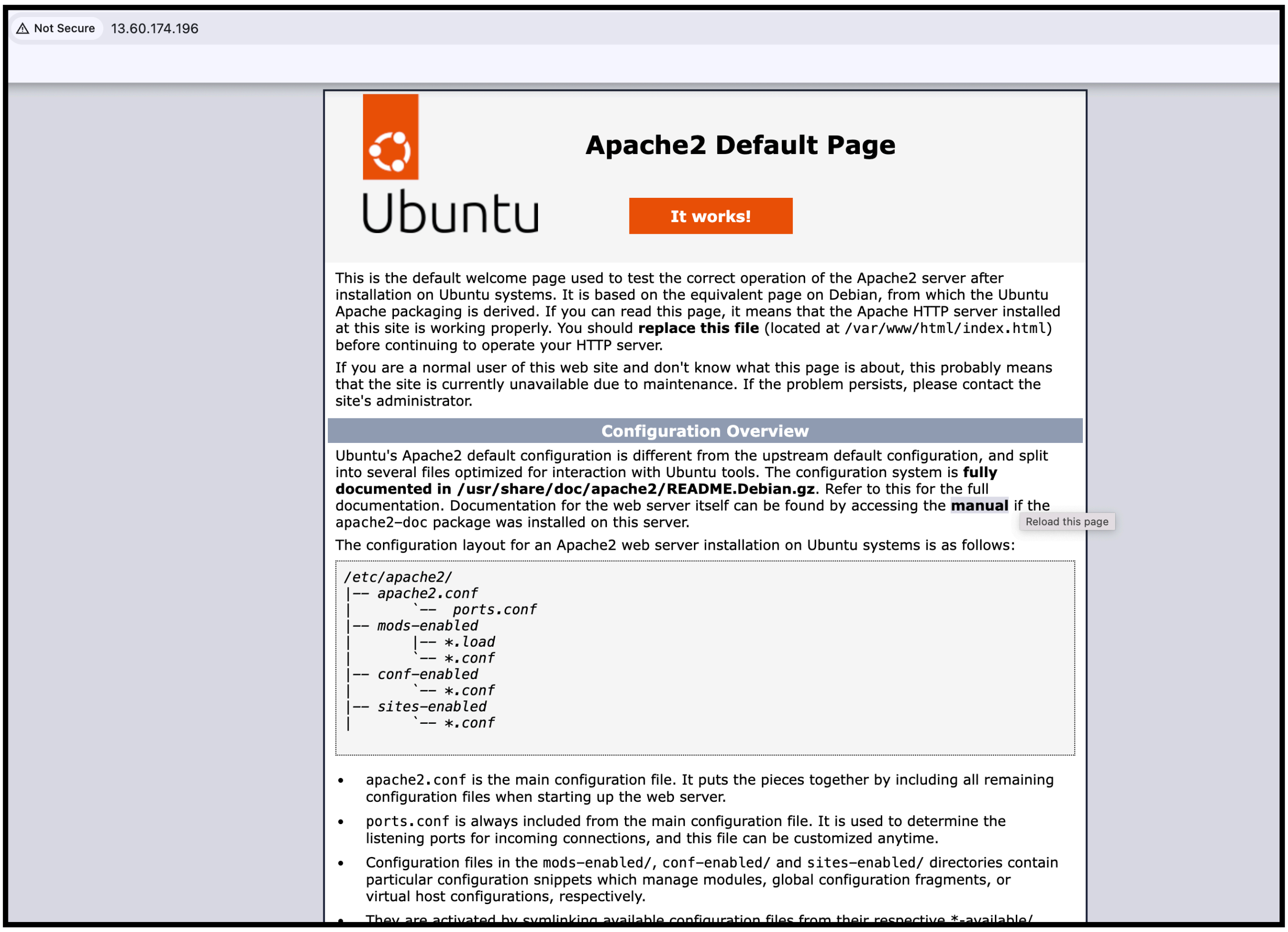
—->worker1

```
apache2: Syntax error on line 80 of /etc/apache2/apache2.conf: DefaultRuntimeDir must be a valid dir
root@ip-172-31-33-34:~# systemctl status apache2
● apache2.service - The Apache HTTP Server
     Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled; preset: enabled)
     Active: active (running) since Wed 2026-01-28 11:04:51 UTC; 5min ago
       Docs: https://httpd.apache.org/docs/2.4/
   Main PID: 4007 (apache2)
      Tasks: 55 (limit: 1008)
     Memory: 5.7M (peak: 6.2M)
        CPU: 53ms
     CGroup: /system.slice/apache2.service
             ├─4007 /usr/sbin/apache2 -k start
             ├─4010 /usr/sbin/apache2 -k start
             └─4011 /usr/sbin/apache2 -k start

Jan 28 11:04:51 ip-172-31-33-34 systemd[1]: Starting apache2.service - The Apache HTTP Server...
Jan 28 11:04:51 ip-172-31-33-34 systemd[1]: Started apache2.service - The Apache HTTP Server.
root@ip-172-31-33-34:~#
```

i-0e57dafb81ac0e460 (worker1)

PublicIPs: 13.60.174.196   PrivateIPs: 172.31.33.34



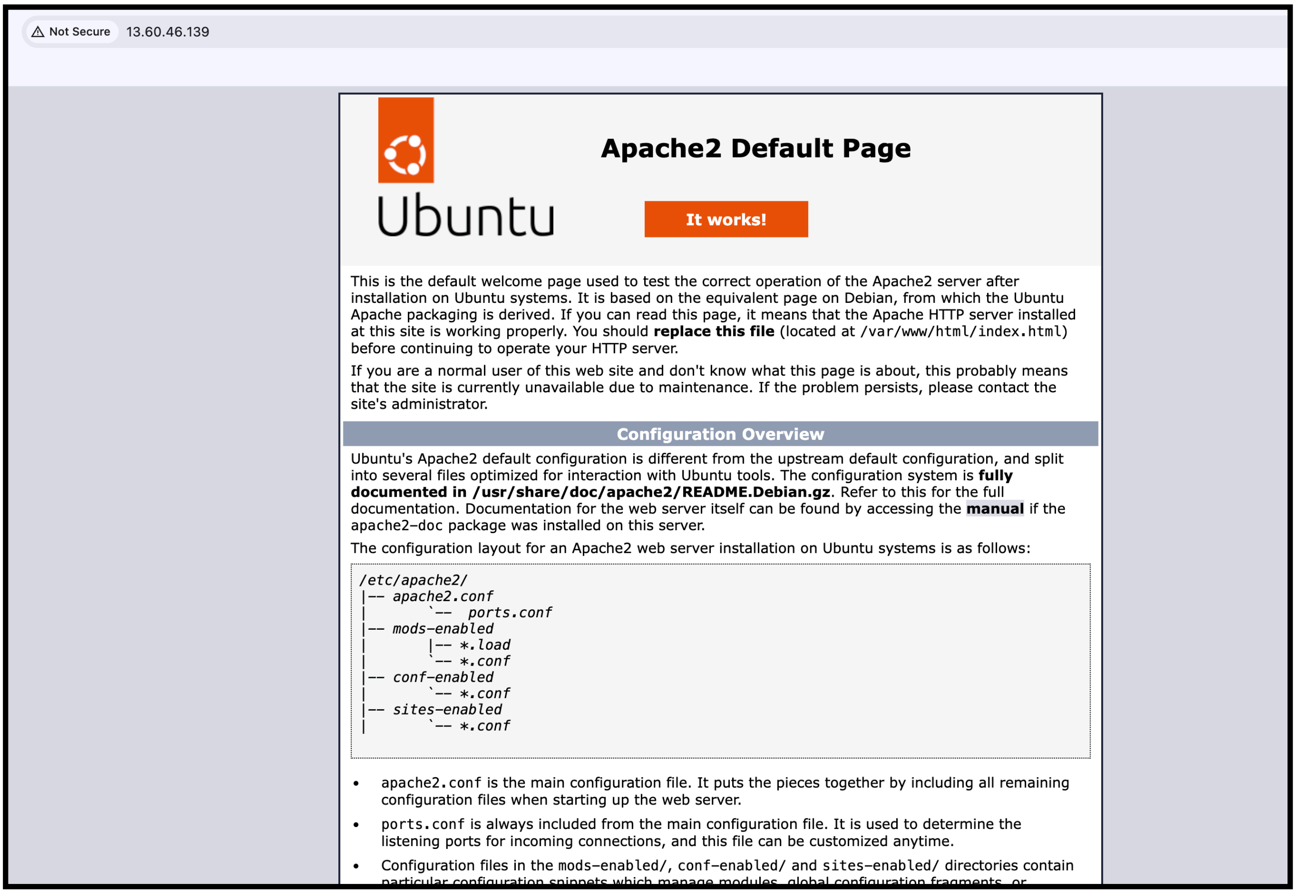
—>worker2

```
root@ip-172-31-37-159:~# systemctl status apache2
● apache2.service - The Apache HTTP Server
     Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled; preset: enabled)
     Active: active (running) since Wed 2026-01-28 11:04:52 UTC; 6min ago
       Docs: https://httpd.apache.org/docs/2.4/
   Main PID: 3906 (apache2)
      Tasks: 55 (limit: 1008)
     Memory: 5.8M (peak: 6.1M)
        CPU: 71ms
     CGroup: /system.slice/apache2.service
             ├─3906 /usr/sbin/apache2 -k start
             ├─3909 /usr/sbin/apache2 -k start
             └─3910 /usr/sbin/apache2 -k start

Jan 28 11:04:52 ip-172-31-37-159 systemd[1]: Starting apache2.service - The Apache HTTP Server...
Jan 28 11:04:52 ip-172-31-37-159 systemd[1]: Started apache2.service - The Apache HTTP Server.
root@ip-172-31-37-159:~#
```

i-09212a024d7a75d98 (worker2)

PublicIPs: 13.60.46.139   PrivateIPs: 172.31.37.159



Start or stop the service
To Start
ansible all -m service -a "name=httpd state=started enabled=yes"
To Stop
ansible all -m service -a "name=httpd state=stop enabled=yes"

To stop
—-> ansible all -m service -a "name=apache2 state=stopped enabled=yes"

```
root@ip-172-31-2-72:/etc/ansible# ansible all -m service -a "name=apache2 state=stopped enabled=yes"
172.31.33.34 | CHANGED => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3"
    },
    "changed": true,
    "enabled": true,
    "name": "apache2",
    "state": "stopped",
    "status": {
        "ActiveEnterTimestamp": "Wed 2026-01-28 11:04:51 UTC",
        "ActiveEnterTimestampMonotonic": "6777812261",
        "ActiveExitTimestampMonotonic": "0",
        "ActiveState": "active",
        "After": "nss-lookup.target basic.target -.mount sysinit.target tmp.mount system.slice systemd-journald.socket network.target remote-fs.targe
        "AllowIsolate": "no",
        "AssertResult": "yes",
        "AssertTimestamp": "Wed 2026-01-28 11:04:51 UTC",
        "AssertTimestampMonotonic": "6777777962",
        "Before": "shutdown.target apache-htcacheclean.service multi-user.target",
        "BlockIOAccounting": "no",
        "BlockIOWeight": "[not set]",
        "CPUAccounting": "yes",
        "CPUAffinityFromNUMA": "no",
        "CPUQuotaPerSecUSec": "infinity",
        "CPUQuotaPeriodUSec": "infinity",
        "CPUSchedulingPolicy": "0",
        "CPUSchedulingPriority": "0",
        "CPUSchedulingResetOnFork": "no",
        "CPUShares": "[not set]",
```

—> check your workers
—> worker1

```
root@ip-172-31-33-34:~# sudo systemctl status apache2
○ apache2.service - The Apache HTTP Server
     Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled; preset: enabled)
     Active: inactive (dead) since Wed 2026-01-28 11:17:05 UTC; 21s ago
   Duration: 12min 13.478s
       Docs: https://httpd.apache.org/docs/2.4/
    Process: 4600 ExecStop=/usr/sbin/apachectl graceful-stop (code=exited, status=0/SUCCESS)
   Main PID: 4007 (code=exited, status=0/SUCCESS)
        CPU: 95ms

Jan 28 11:04:51 ip-172-31-33-34 systemd[1]: Starting apache2.service - The Apache HTTP Server...
Jan 28 11:04:51 ip-172-31-33-34 systemd[1]: Started apache2.service - The Apache HTTP Server.
Jan 28 11:17:05 ip-172-31-33-34 systemd[1]: Stopping apache2.service - The Apache HTTP Server...
Jan 28 11:17:05 ip-172-31-33-34 systemd[1]: apache2.service: Deactivated successfully.
Jan 28 11:17:05 ip-172-31-33-34 systemd[1]: Stopped apache2.service - The Apache HTTP Server.
root@ip-172-31-33-34:~#
```

i-0e57dafb81ac0e460 (worker1)

PublicIPs: 13.60.174.196 PrivateIPs: 172.31.33.34

—> the service is stopped

—> similarly even worker2 service is also stopped

Create the user on ALL servers

—> ansible all -m user -a "name=weblogic state=present"

```
root@ip-172-31-2-72:/etc/ansible# ansible all -m user -a "name=weblogic state=present"
172.31.33.34 | CHANGED => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3"
    },
    "changed": true,
    "comment": "",
    "create_home": true,
    "group": 1001,
    "home": "/home/weblogic",
    "name": "weblogic",
    "shell": "/bin/sh",
    "state": "present",
    "system": false,
    "uid": 1001
}
172.31.37.159 | CHANGED => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3"
    },
    "changed": true,
    "comment": "",
    "create_home": true,
    "group": 1001,
    "home": "/home/weblogic",
    "name": "weblogic",
    "shell": "/bin/sh",
    "state": "present",
    "system": false,
    "uid": 1001
}
root@ip-172-31-2-72:/etc/ansible#
```

Verify user exists

ansible all -m shell -a "id weblogic"

```
root@ip-172-31-2-72:/etc/ansible# ansible all -m shell -a "id weblogic"
172.31.37.159 | CHANGED | rc=0 >>
uid=1001(weblogic) gid=1001(weblogic) groups=1001(weblogic)
172.31.33.34 | CHANGED | rc=0 >>
uid=1001(weblogic) gid=1001(weblogic) groups=1001(weblogic)
root@ip-172-31-2-72:/etc/ansible#
```

i-08544442c9a14b500 (ansible-master)

PublicIPs: 16.16.94.111 PrivateIPs: 172.31.2.72

Create a file with 755 permission
ansible all -m file -a "path=/root/testfile state=touch mode=0755"

```
root@ip-172-31-2-72:/etc/ansible# ansible all -m file -a "path=/root/testfile state=touch mode=0755"
172.31.33.34 | CHANGED => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3"
    },
    "changed": true,
    "dest": "/root/testfile",
    "gid": 0,
    "group": "root",
    "mode": "0755",
    "owner": "root",
    "size": 0,
    "state": "file",
    "uid": 0
}
172.31.37.159 | CHANGED => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3"
    },
    "changed": true,
    "dest": "/root/testfile",
    "gid": 0,
    "group": "root",
    "mode": "0755",
    "owner": "root",
    "size": 0,
    "state": "file",
    "uid": 0
}
root@ip-172-31-2-72:/etc/ansible#
```

```
-rwxr-xr-x   1 root root     0 Jan 28 11:34 testfile*
root@ip-172-31-33-34:~#
```

i-0e57dafb81ac0e460 (worker1)

PublicIPs: 13.60.174.196   PrivateIPs: 172.31.33.34

Change ownership of a file
ansible all -m file -a "path=/root/testfile group=weblogic owner=weblogic" -b

Tips: -b is the option for become and by default it will become root user

–K is to tell ansible to ask for SUDO password

```
root@ip-172-31-2-72:/etc/ansible# ansible all -m file -a "path=/root/testfile group=weblogic owner=weblogic" -b
172.31.33.34 | CHANGED => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3"
    },
    "changed": true,
    "gid": 1001,
    "group": "weblogic",
    "mode": "0755",
    "owner": "weblogic",
    "path": "/root/testfile",
    "size": 0,
    "state": "file",
    "uid": 1001
}
172.31.37.159 | CHANGED => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3"
    },
    "changed": true,
    "gid": 1001,
    "group": "weblogic",
    "mode": "0755",
    "owner": "weblogic",
    "path": "/root/testfile",
    "size": 0,
    "state": "file",
    "uid": 1001
}
root@ip-172-31-2-72:/etc/ansible#
```

```
-rwxr-xr-x  1 weblogic weblogic    0 Jan 28 11:34 testfile*
root@ip-172-31-37-159:~#
```

i-09212a024d7a75d98 (worker2)

PublicIPs: 13.60.46.139   PrivateIPs: 172.31.37.159

CloudShell   Feedback   Console Mobile App

Create a Linux user group
ansible all -m group -a "name=test state=present" -b -K

```
root@ip-172-31-2-72:/etc/ansible# ansible all -m group -a "name=test state=present" -b -K
BECOME password:
172.31.33.34 | CHANGED => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3"
    },
    "changed": true,
    "gid": 1002,
    "name": "test",
    "state": "present",
    "system": false
}
172.31.37.159 | CHANGED => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3"
    },
    "changed": true,
    "gid": 1002,
    "name": "test",
    "state": "present",
    "system": false
}
root@ip-172-31-2-72:/etc/ansible#
```

i-08544442c9a14b500 (ansible-master)

PublicIPs: 16.16.94.111   PrivateIPs: 172.31.2.72